

Magic Bitstrings

Tags:

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AI Code Tips

Description

A bitstring, whose length is one less than a prime, might be **magic**. 1001 is one such string. In order to see the **magic** in the string let us append a non-bit $\times$ to it, regard the new *thingy* as a cyclic string, and make this square matrix of bits

each bit	1001
every 2 nd bit	0110
every 3 rd bit	0110
every 4 th bit	1001

This matrix has the same number of rows as the length of the original bitstring. The m -th row of the matrix has every m -th bit of the original string starting with the m -th bit. Because the enlarged *thingy* has prime length, the appended $\times$ never gets used.
 If each row of the matrix is either the original bitstring or its complement, the original bitstring is **magic**.

Each line of input (except last) contains a prime number $p \leq 100000$. The last line contains 0 and this line should not be processed. For each prime number from the input produce one line of output containing the lexicographically smallest, non-constant **magic** bitstring of length $p-1$, if such a string exists, otherwise output *Impossible*.

Samples

```

5
3
17
47
2
79
0

```

```

0110
01
0010111001110100
00001000011010100011011001001111010111
Impossible
001001100001011010000001001111001110101010001100001101111101001011110011011

```

Submit

Codes

[HZNUOJ](#) is based on [HUSTOJ](#)

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